



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

Instructor: Prashant Kumar Pandey



Azure
Databricks

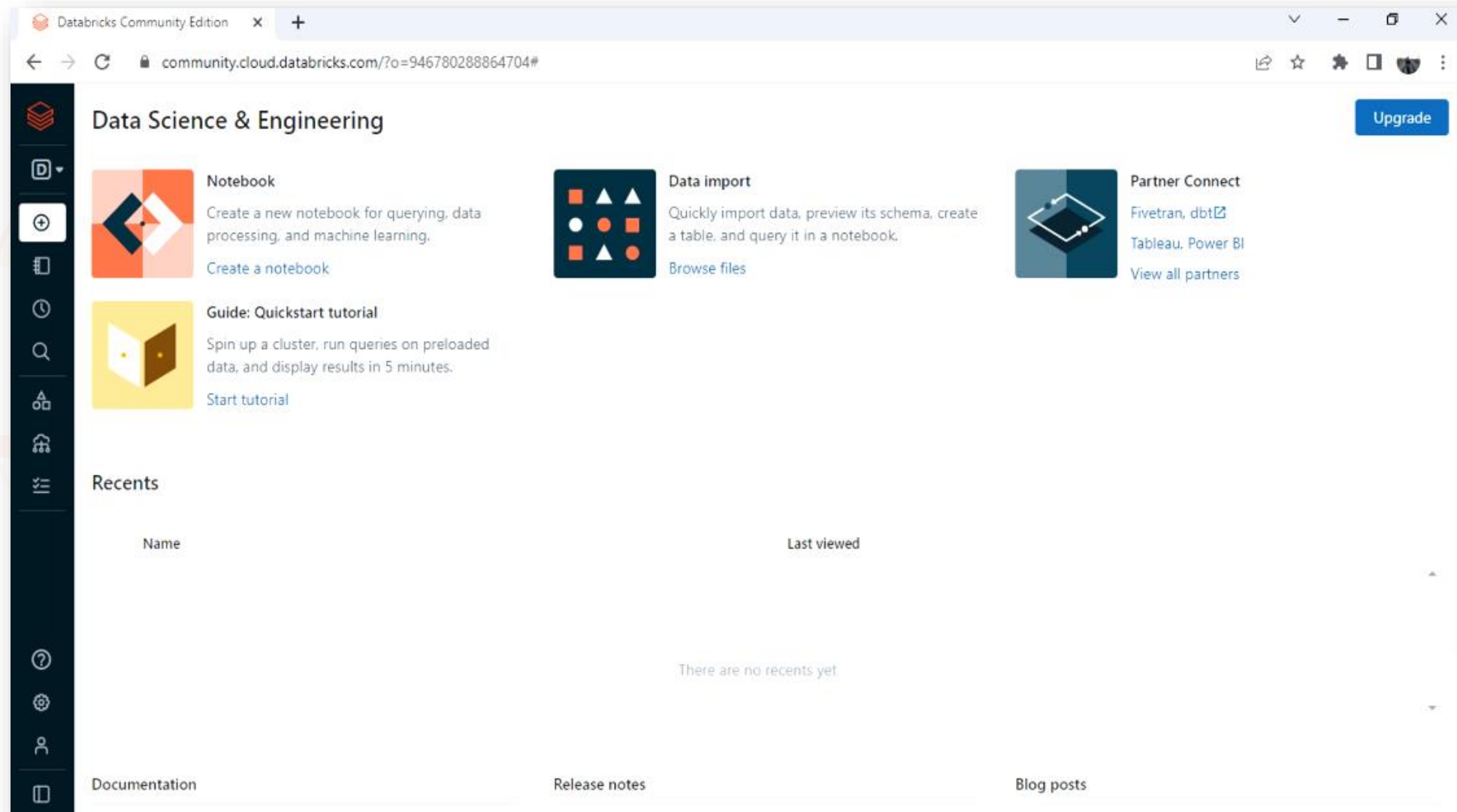


Absolute Beginner to Specialization in Apache Spark and Azure Databricks

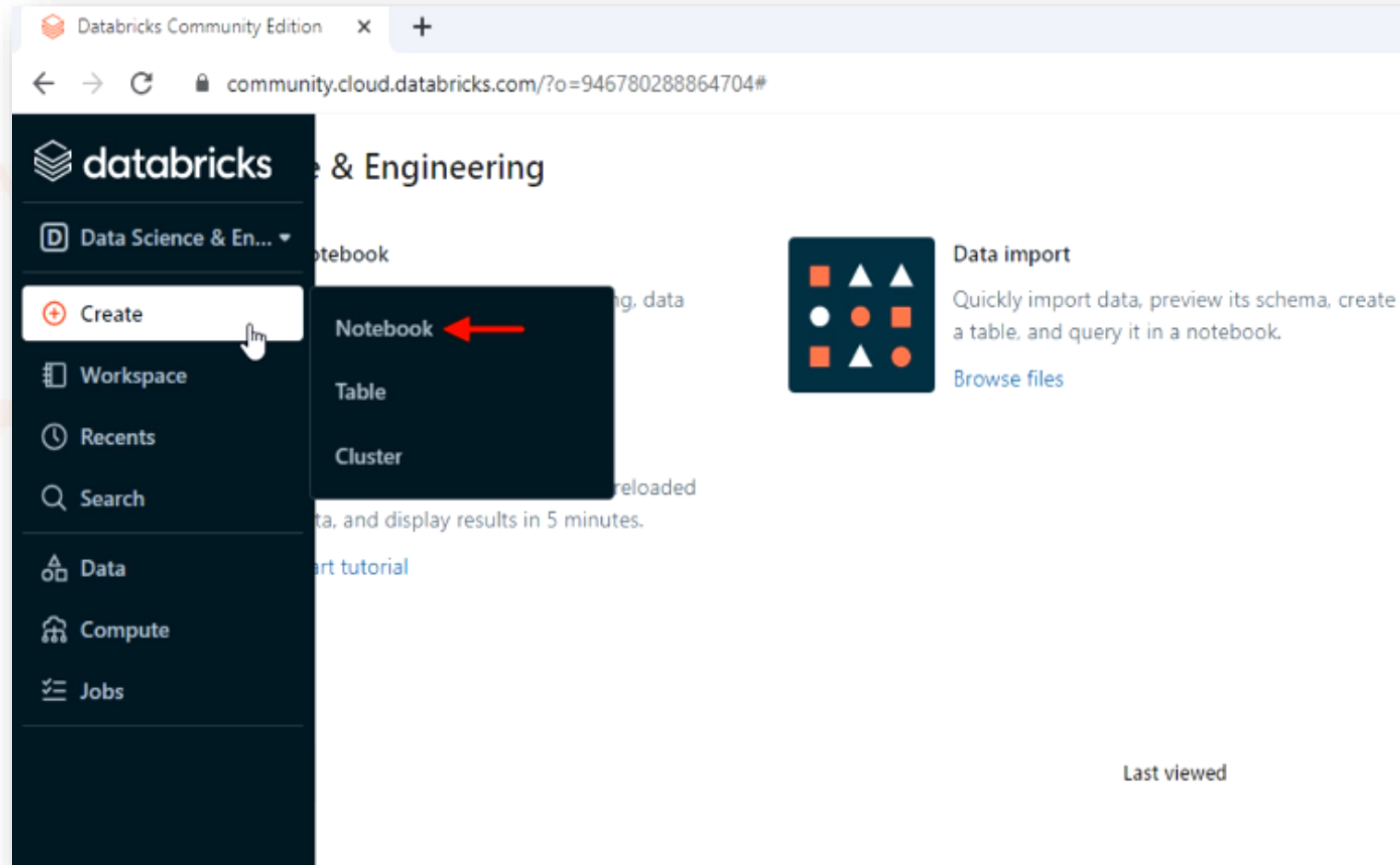


First Spark Application in Databricks Cloud

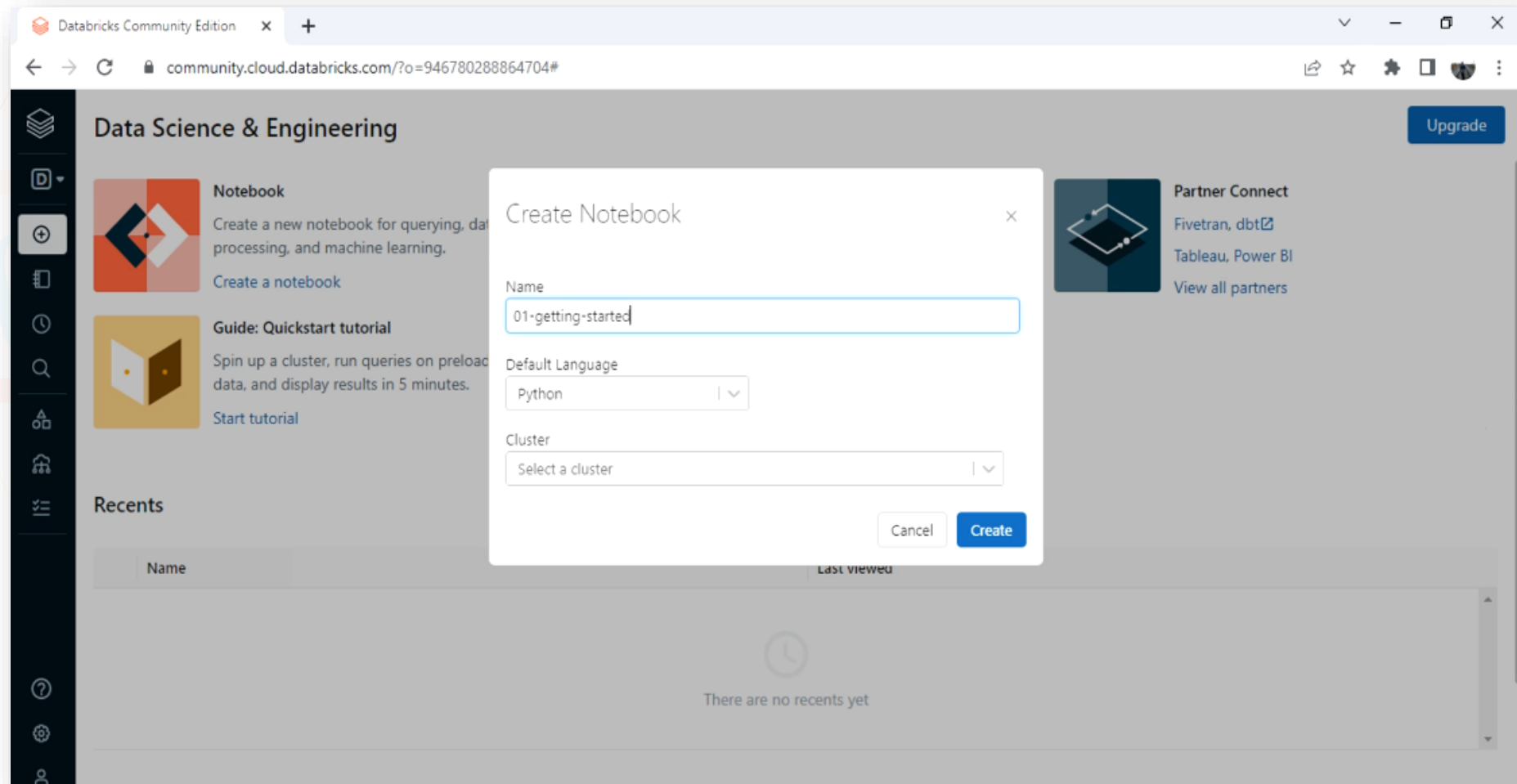
Login to your Databricks Workspace.



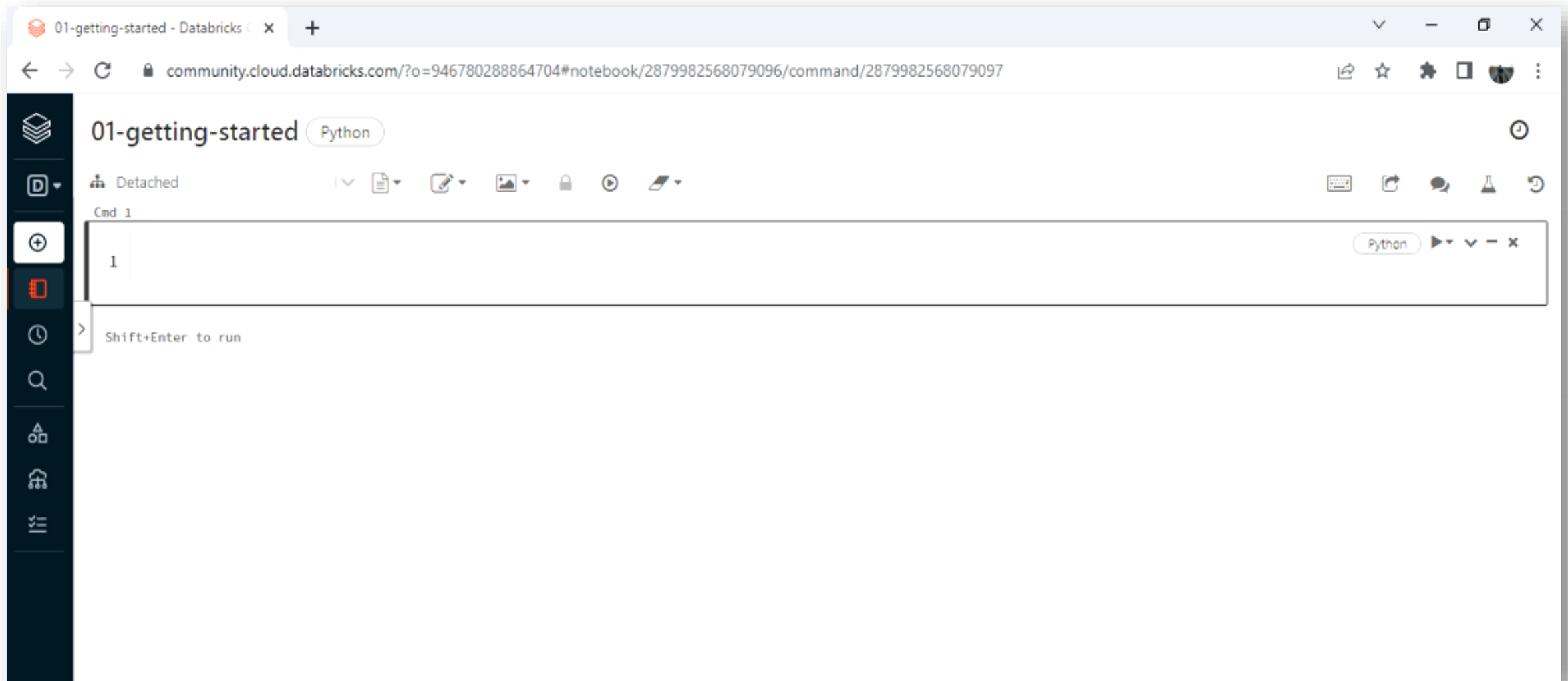
Come to the navigation menu and hit the create button. You will see three options there: Notebook, Table and Cluster. Hit the notebook option and start creating a notebook.



Give a name to your notebook and hit the create button to get started.



Here is your notebook.



Requirements:

We have some data available in the cloud environment at the location given below. It is a CSV format data file, which contains some information about the diamonds.

Requirement

Given data file

```
/databricks-datasets/Rdatasets/data-001/csv/ggplot2/diamonds.csv
```

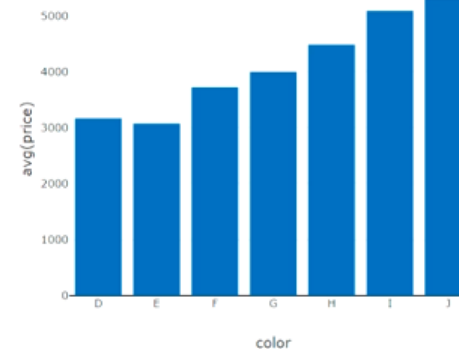
1. Read show

	_c0 carat	cut color	clarity depth	table price	x	y	z
1	0.23	Ideal	E	SI2 61.5 55.0	326 3.95 3.98	2.43	
2	0.21	Premium	E	SI1 59.8 61.0	326 3.89 3.84	2.31	
3	0.23	Good	E	VS1 56.9 65.0	327 4.05 4.07	2.31	
4	0.29	Premium	I	VS2 62.4 58.0	334 4.2 4.23	2.63	
5	0.31	Good	J	SI2 63.3 58.0	335 4.34 4.35	2.75	

2. Average price by colour.

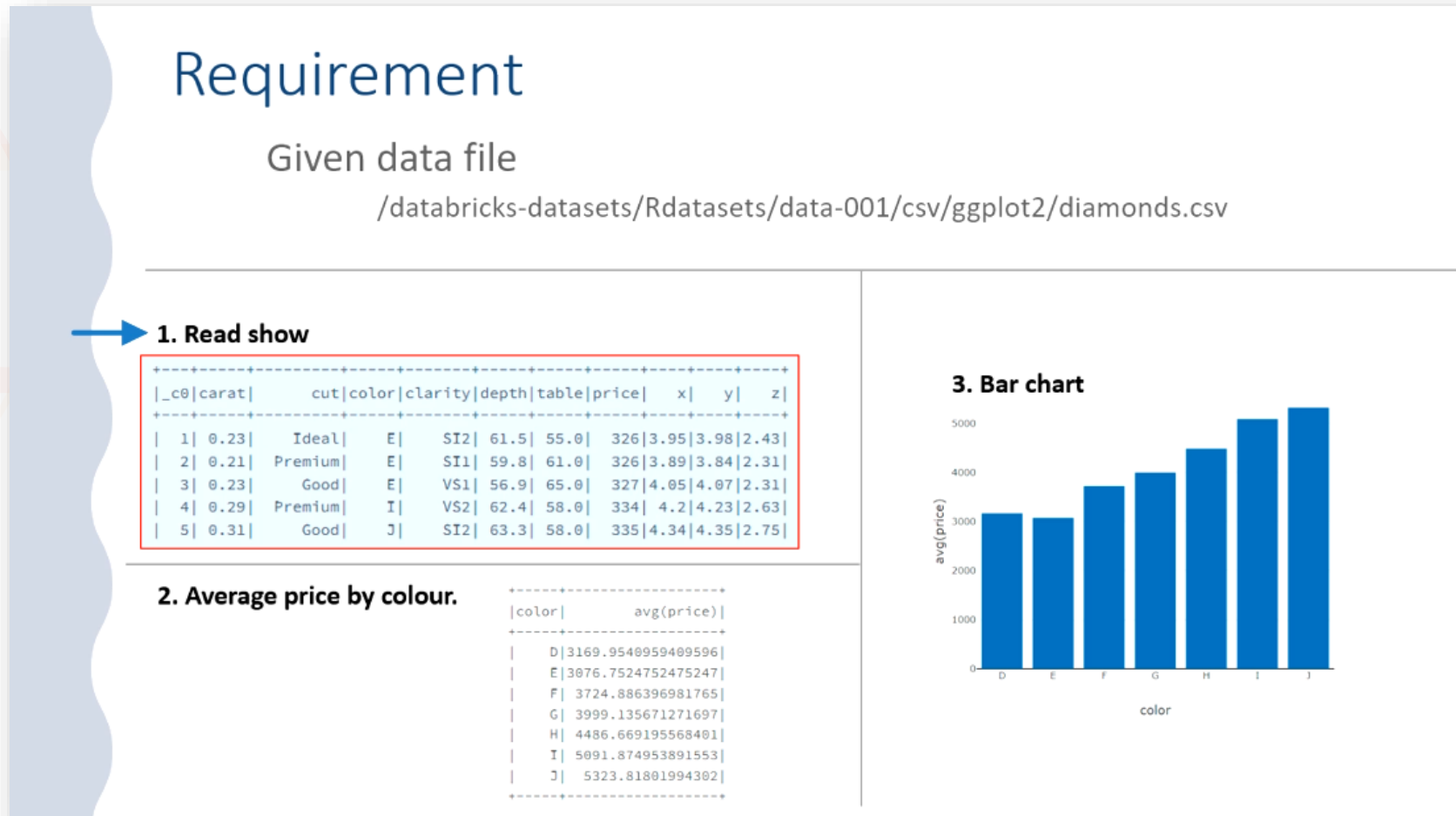
color	avg(price)
D	3169.9540959409596
E	3076.7524752475247
F	3724.886396981765
G	3999.135671271697
H	4486.669195568401
I	5091.874953891553
J	5323.81801994302

3. Bar chart



Requirements:

1. Read the data file and show some records in tabular format. And the output should look like the image highlighted below.



Requirements:

2. Calculate the Average price of the diamonds by color. And the result might look like the image highlighted below.

Requirement

Given data file

/databricks-datasets/Rdatasets/data-001/csv/ggplot2/diamonds.csv

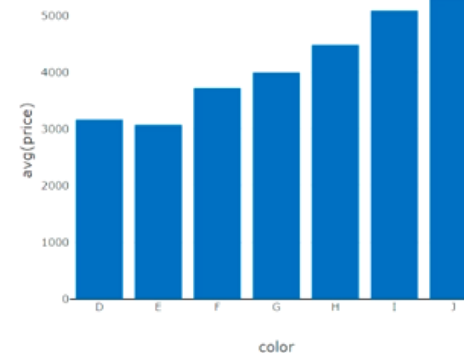
1. Read show

_c0 carat	cut color clarity depth table price	x	y	z
1 0.23	Ideal E SI2 61.5 55.0 326 3.95 3.98 2.43			
2 0.21	Premium E SI1 59.8 61.0 326 3.89 3.84 2.31			
3 0.23	Good E VS1 56.9 65.0 327 4.05 4.07 2.31			
4 0.29	Premium I VS2 62.4 58.0 334 4.2 4.23 2.63			
5 0.31	Good J SI2 63.3 58.0 335 4.34 4.35 2.75			

2. Average price by colour.

color	avg(price)
D	3169.9540959409596
E	3076.7524752475247
F	3724.886396981765
G	3999.135671271697
H	4486.669195568401
I	5091.874953891553
J	5323.81801994302

3. Bar chart



Requirements:

3. Draw a bar chart using this table to visualize the results. And the bar chart might look like the image highlighted below.

Requirement

Given data file

/databricks-datasets/Rdatasets/data-001/csv/ggplot2/diamonds.csv

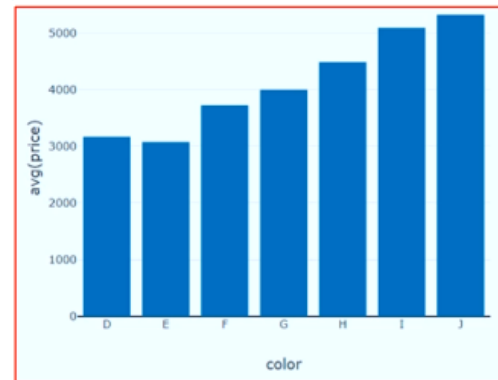
1. Read show

_c0	carat	cut	color	clarity	depth	table	price	x	y	z
1	0.23	Ideal	E	SI2	61.5	55.0	326	3.95	3.98	2.43
2	0.21	Premium	E	SI1	59.8	61.0	326	3.89	3.84	2.31
3	0.23	Good	E	VS1	56.9	65.0	327	4.05	4.07	2.31
4	0.29	Premium	I	VS2	62.4	58.0	334	4.2	4.23	2.63
5	0.31	Good	J	SI2	63.3	58.0	335	4.34	4.35	2.75

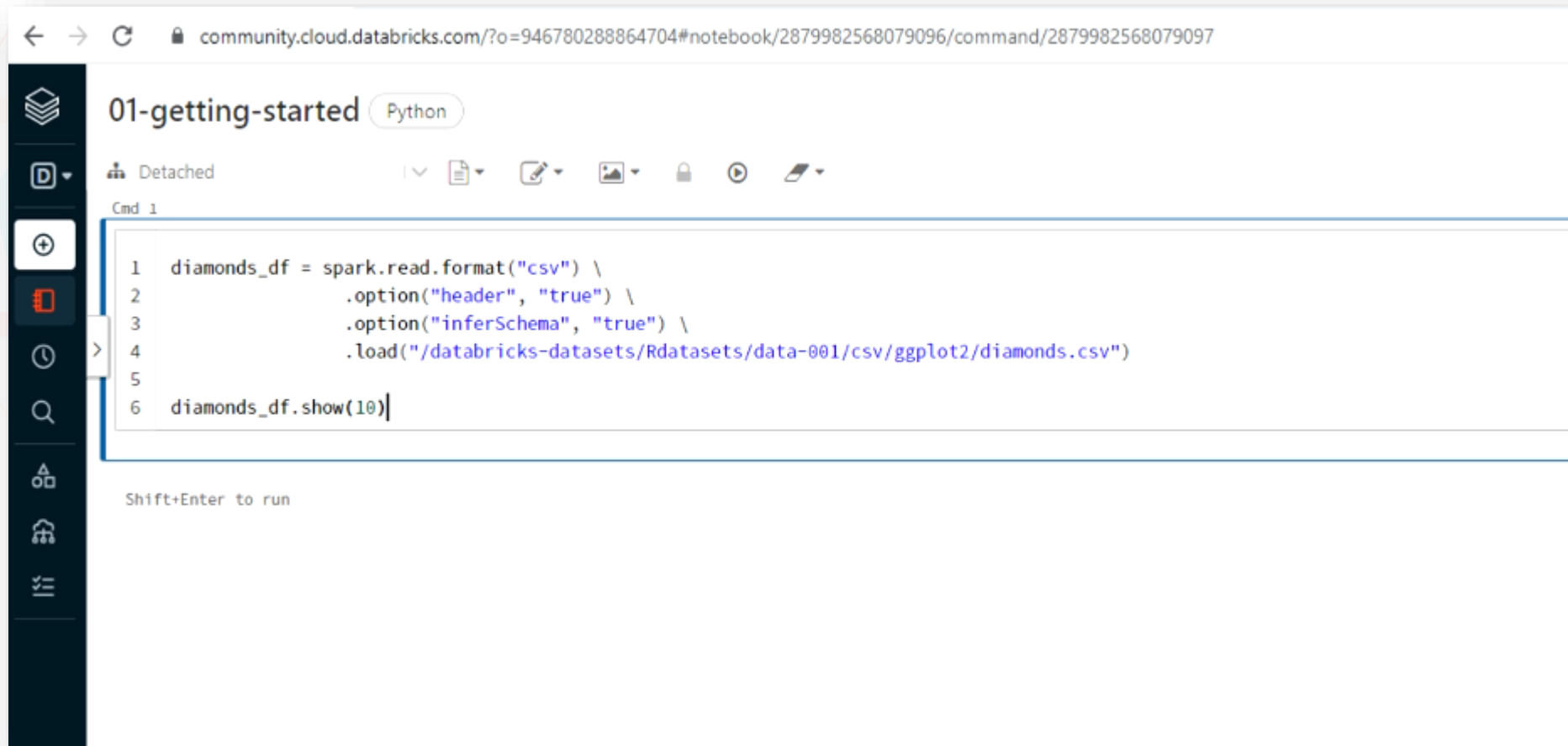
2. Average price by colour.

color	avg(price)
D	3169.9540959409596
E	3076.7524752475247
F	3724.886396981765
G	3999.135671271697
H	4486.669195568401
I	5091.874953891553
J	5323.81801994302

3. Bar chart



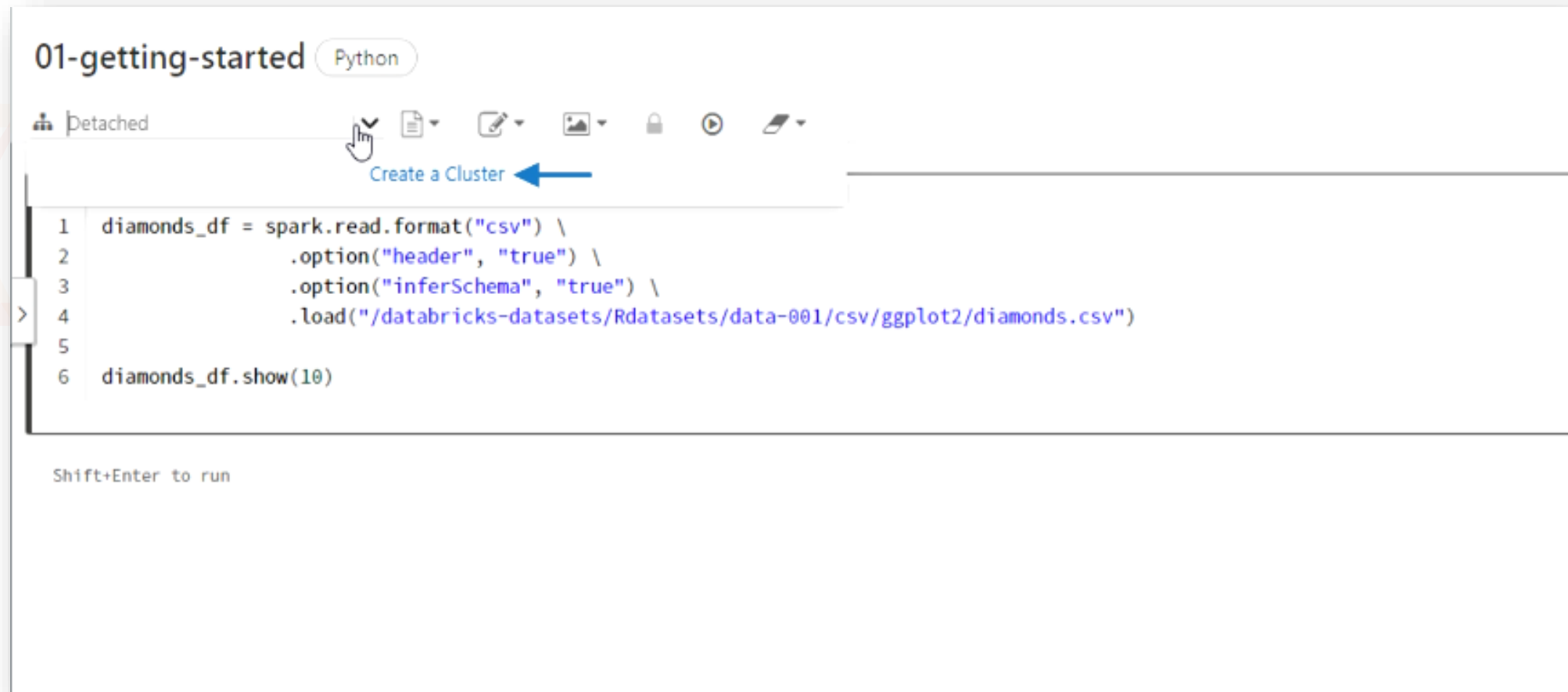
Here is the code shown below which meets your first requirement.
(Reference: 01-getting-started.ipynb)
It will read the data file and show some records in tabular format.

A screenshot of a Databricks notebook interface. The browser address bar shows the URL: community.cloud.databricks.com/?o=946780288864704#notebook/2879982568079096/command/2879982568079097. The notebook title is "01-getting-started" with a "Python" language selector. The interface includes a sidebar with icons for workspace, recent notebooks, search, and other functions. The main area shows a code cell labeled "Cmd 1" containing the following Python code:

```
1 diamonds_df = spark.read.format("csv") \  
2     .option("header", "true") \  
3     .option("inferSchema", "true") \  
4     .load("/databricks-datasets/Rdatasets/data-001/csv/ggplot2/diamonds.csv") \  
5 \  
6 diamonds_df.show(10)
```

Below the code cell, there is a prompt "Shift+Enter to run".

Now, we want to run this code and see the results. But running a Spark code requires a Spark cluster. As shown in the image below, click the dropdown menu, and you will see a link to create a cluster. Hit the link, and it will take you to the cluster creation page.



Give a name to the cluster and hit the create button, it might take a few minutes to wait for the cluster to start.

Create Cluster - Databricks Com x +

community.cloud.databricks.com/?o=946780288864704#create/cluster

Create Cluster

New Cluster Cancel Create Cluster **0 Workers:0 GB Memory, 0 Cores, 0 DBU**
1 Driver:15.3 GB Memory, 2 Cores, 1 DBU

Cluster name

Databricks runtime version ⓘ

Runtime: 9.1 LTS (Scala 2.12, Spark 3.1.2) | v

Instance

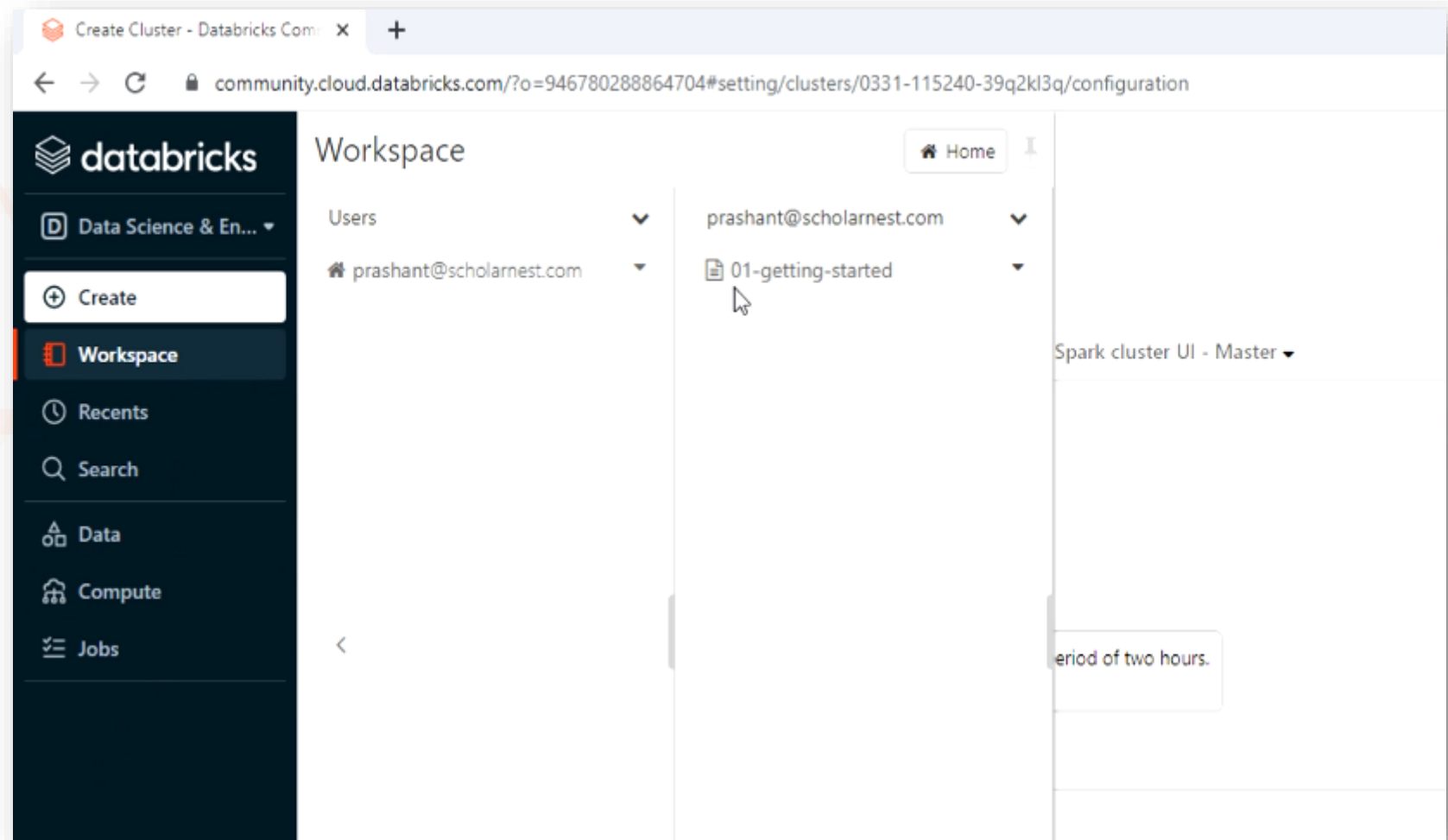
Free 15 GB Memory: As a Community Edition user, your cluster will automatically terminate after an idle period of two hours. For [more configuration options](#), please upgrade your Databricks subscription.

Instances Spark

Availability zone ⓘ

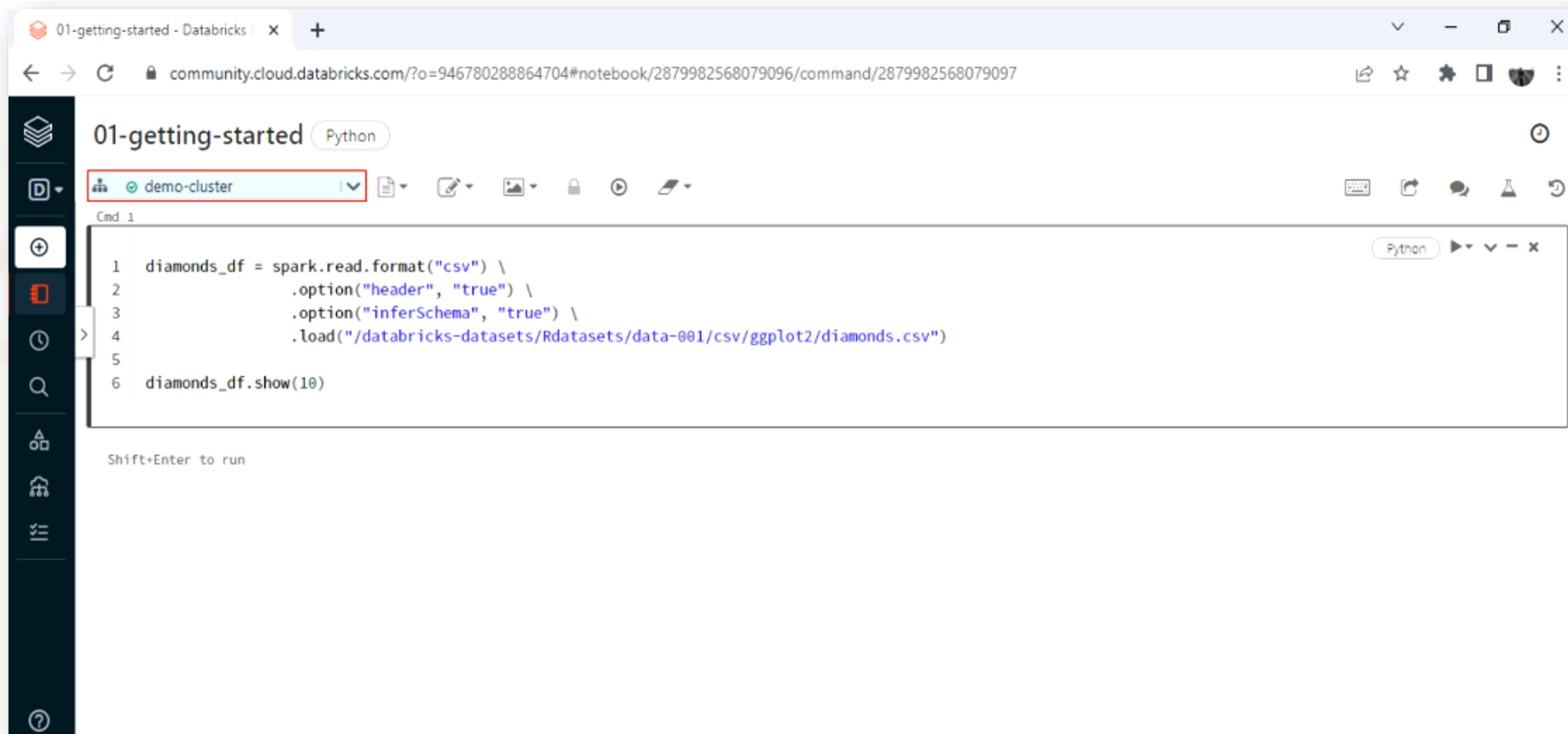
auto | v

Now, go to the workspace menu option and go back to your notebook.



Make sure your cluster is attached to the notebook, and your code will run on this cluster.

Now, click inside the code cell and press shift+enter to run the code.



You can see the results here, and it matches our expected results. So we loaded the diamonds.csv file into Spark and showed ten records from the data file.

```
01-getting-started Python
demo-cluster
load("/databricks-datasets/Rdatasets/data-001/csv/ggplot2/diamonds.csv")
diamonds_df.show(10)
```

(3) Spark Jobs

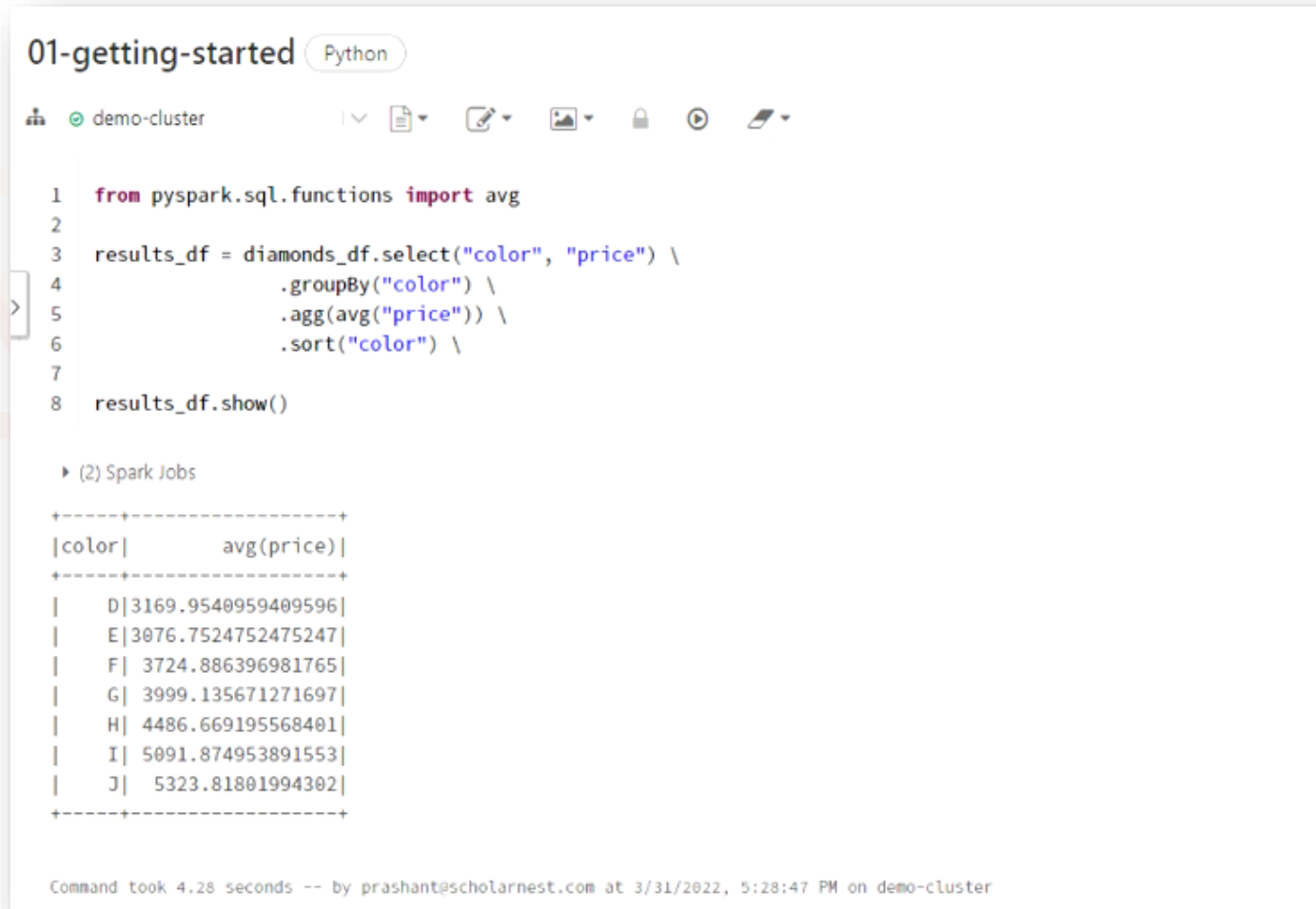
_c0	carat	cut	color	clarity	depth	table	price	x	y	z
1	0.23	Ideal	E	SI2	61.5	55.0	326	3.95	3.98	2.43
2	0.21	Premium	E	SI1	59.8	61.0	326	3.89	3.84	2.31
3	0.23	Good	E	VS1	56.9	65.0	327	4.05	4.07	2.31
4	0.29	Premium	I	VS2	62.4	58.0	334	4.2	4.23	2.63
5	0.31	Good	J	SI2	63.3	58.0	335	4.34	4.35	2.75
6	0.24	Very Good	J	VVS2	62.8	57.0	336	3.94	3.96	2.48
7	0.24	Very Good	I	VVS1	62.3	57.0	336	3.95	3.98	2.47
8	0.26	Very Good	H	SI1	61.9	55.0	337	4.07	4.11	2.53
9	0.22	Fair	E	VS2	65.1	61.0	337	3.87	3.78	2.49
10	0.23	Very Good	H	VS1	59.4	61.0	338	4.0	4.05	2.39

only showing top 10 rows

Command took 13.09 seconds -- by prashant@scholarnest.com at 3/31/2022, 5:24:52 PM on demo-cluster

The following requirement is to process this diamond's data frame and calculate the average price by color. And here is the code for the same.

So this code takes color and price columns, groups them by color, and calculates the average price. Finally, I show the results. You can also see the output shown below.



```
01-getting-started Python
demo-cluster

1 from pyspark.sql.functions import avg
2
3 results_df = diamonds_df.select("color", "price") \
4     .groupBy("color") \
5     .agg(avg("price")) \
6     .sort("color") \
7
8 results_df.show()
```

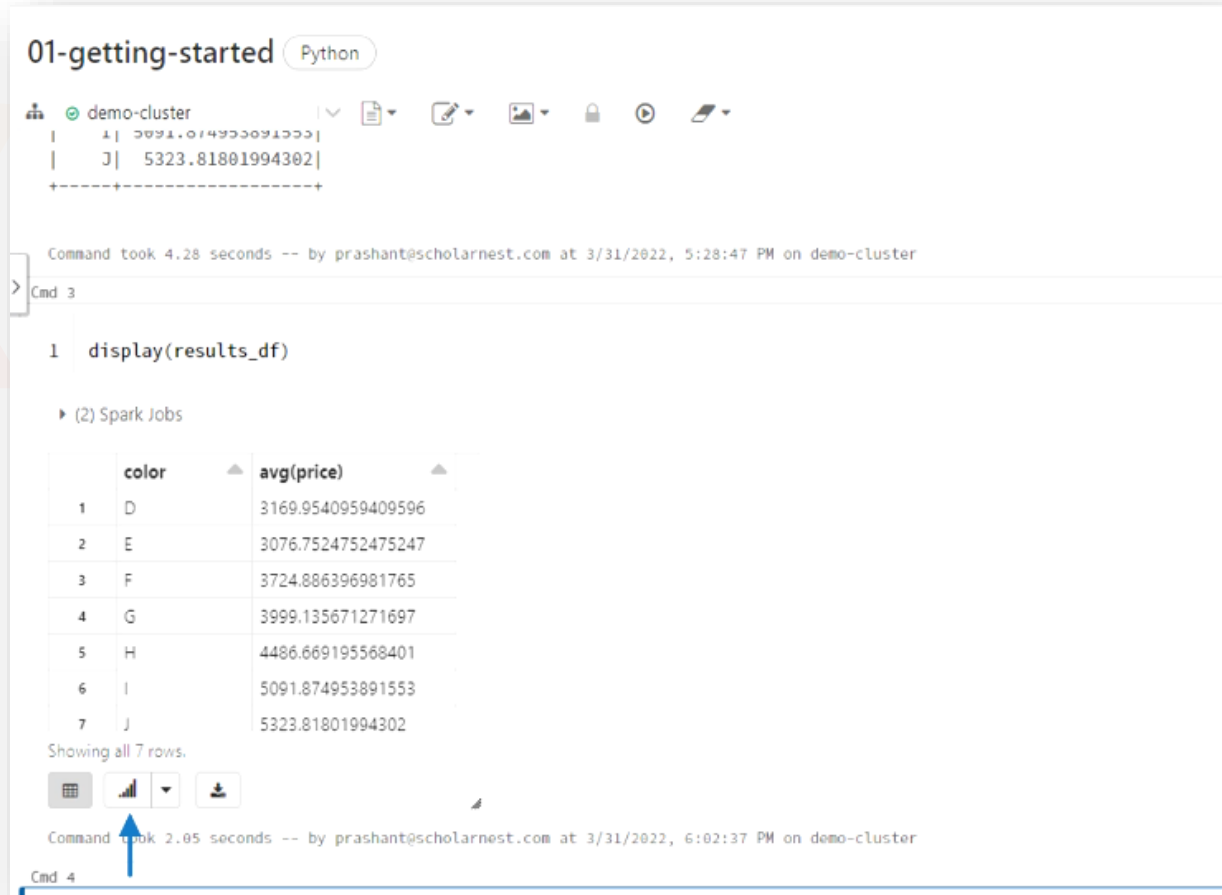
▶ (2) Spark Jobs

color	avg(price)
D	3169.9540959409596
E	3076.7524752475247
F	3724.886396981765
G	3999.135671271697
H	4486.669195568401
I	5091.874953891553
J	5323.81801994302

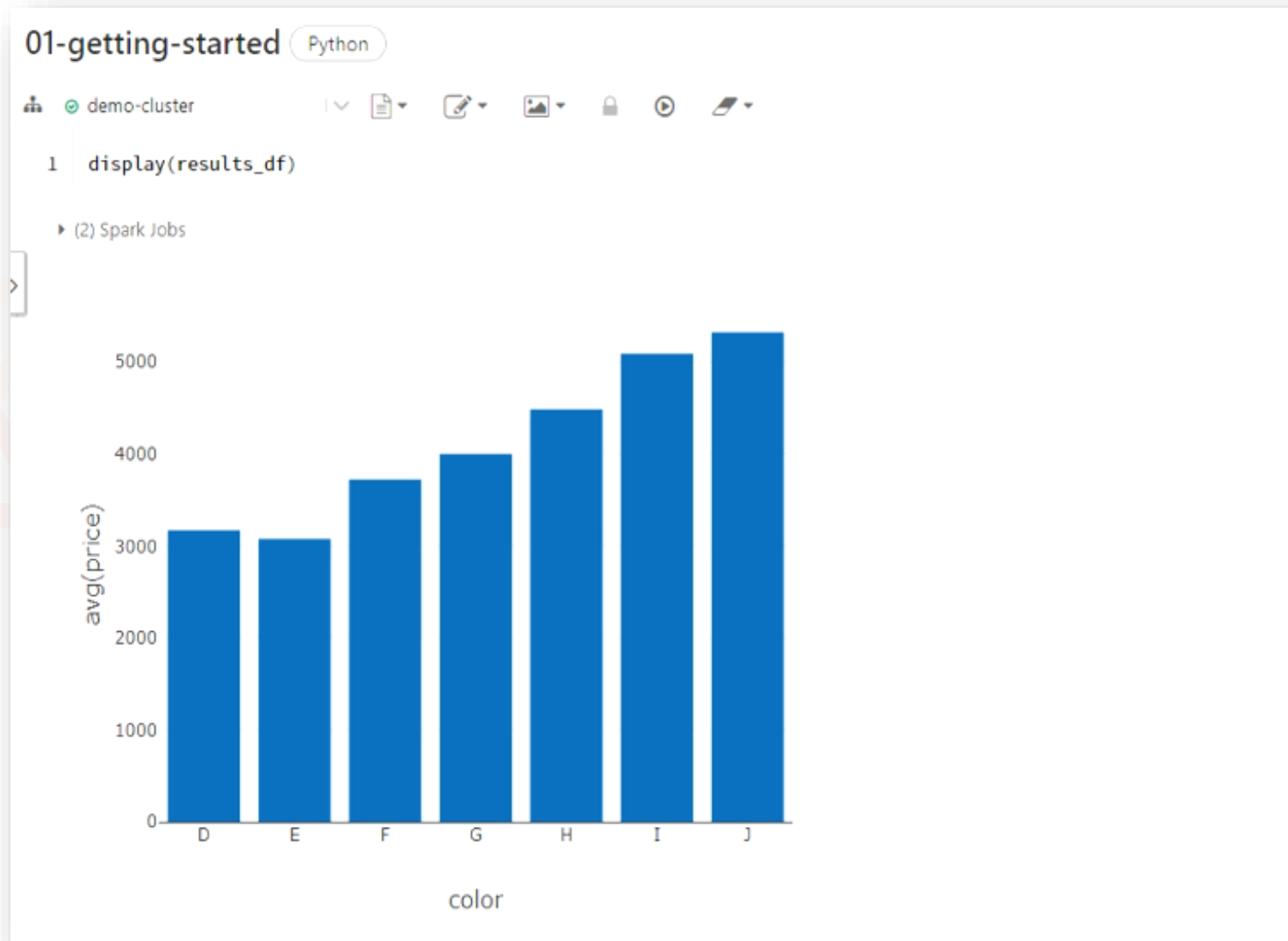
Command took 4.28 seconds -- by prashant@scholarnest.com at 3/31/2022, 5:28:47 PM on demo-cluster

In the last requirement, we wanted to present the result obtained using a bar chart. Here is the code for the same. And you can see the output below.

The display function allows you to format the results. The display function depicts the result in the tabular format but you can switch to the chart format by clicking chart button below the results.



And here is the bar chart we wanted.





Thank You
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